

MATRICES AND MATRIX-VECTOR MULTIPLICATION

A Detailed Foundation for Quantum Computing

From matrix structure and linear transformations to gate composition,
adjoints, inverses, basis changes, and multi-qubit operators

Quantum Mathematics Roadmap - Topic 3

Prepared as a standalone reference volume

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How to use this volume: Sections 1-12 build the linear-algebra foundation. Sections 13-14 translate that foundation into quantum gates and multi-qubit operations. Work through the examples before checking the solutions.

Learning objectives

- Read matrix dimensions and entry notation correctly.
- Compute matrix-vector and matrix-matrix products with confidence.
- Interpret a matrix through its action on basis vectors and the whole space.
- Understand composition order, inverses, determinants, rank, and null spaces.
- Construct adjoints and recognize their role in quantum mechanics.
- Connect one-qubit and multi-qubit gates to concrete matrices.

1. Why matrices are the language of quantum operations

A quantum state is represented by a complex vector. A quantum operation must therefore be a rule that accepts a state vector and produces another state vector. Matrices provide the natural representation of such linear rules.

$$|\psi_{\text{out}}\rangle = A |\psi_{\text{in}}\rangle$$

For one qubit, the state vector has two components, so a general one-qubit operation is represented by a 2×2 matrix. For two qubits, the state vector has four components, so a general two-qubit operation is represented by a 4×4 matrix. In general, an n -qubit operator acts on a 2^n -dimensional state space and is represented by a $2^n \times 2^n$ matrix.

Core interpretation: State vectors describe what the system is. Matrices describe what is done to the system.

System	State dimension	General operator size
1 qubit	2	2×2
2 qubits	4	4×4
3 qubits	8	8×8
n qubits	2^n	$2^n \times 2^n$

Linearity is not an optional convenience. It is the structural principle that allows a matrix to be determined by its action on a basis and allows quantum amplitudes to interfere consistently.

2. Matrix notation and dimensions

A matrix is a rectangular array of scalars. The scalars may be real or complex. An $m \times n$ matrix has m rows and n columns.

Matrix anatomy: dimensions, rows, columns, and entries

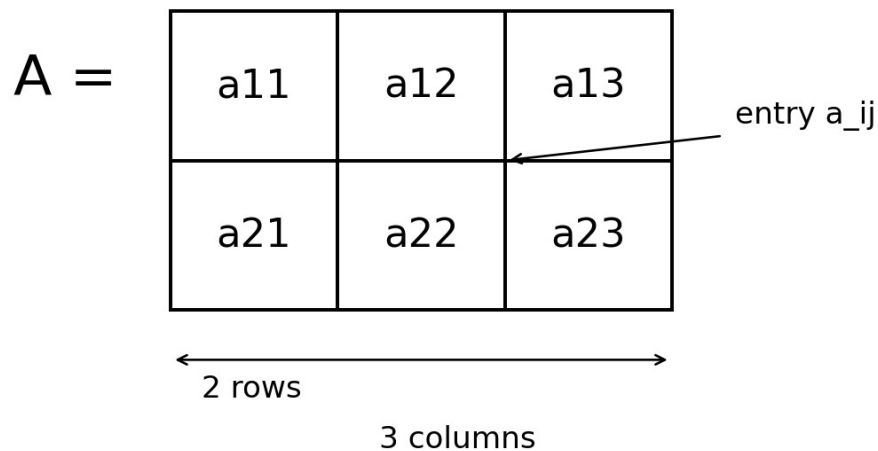


Figure 1. A 2×3 matrix and the meaning of the entry index a_{ij} .

$$A = [a_{ij}], \quad i = \text{row index}, \quad j = \text{column index}$$

For example, a_{23} means the entry in row 2 and column 3. Matrix dimensions are always written as rows $\times$ columns. This convention is essential when checking whether products are defined.

Rows and columns as vectors

Each row can be viewed as a row vector, and each column can be viewed as a column vector. Many matrix properties can be expressed in terms of these vectors. For example, matrix-vector multiplication combines each row with the input vector, while a matrix itself can be understood as a collection of output columns.

Equality of matrices

Two matrices are equal only when they have the same dimensions and every corresponding entry is equal. A 2×3 matrix can never equal a 3×2 matrix, even if the same six numbers appear in both.

$$A = B \quad \text{iff} \quad a_{ij} = b_{ij} \quad \text{for every valid } i, j$$

3. Matrix-vector multiplication

Let A be an $m \times n$ matrix and v an n -component column vector. The product Av is defined and produces an m -component column vector. The inner dimensions must match.

$$(m \times n)(n \times 1) \rightarrow (m \times 1)$$

Matrix-vector multiplication: row-by-column rule

$$\begin{array}{rcl}
 A = [[a, b], [c, d]] & & \begin{array}{l} \text{first row dot } v \\ \swarrow \\ [a x + b y] \end{array} \\
 v = [x, y]^T & A v = & \begin{array}{l} [c x + d y] \\ \nwarrow \\ \text{second row dot } v \end{array}
 \end{array}$$

Each output component is one row of A combined with the whole input vector.

Figure 2. Each output component is the dot product of one row of A with the input vector.

Worked example

$$A = [[2, 1], [-1, 3]], \quad v = [4, 2]^T$$

$$Av = [2(4)+1(2), -1(4)+3(2)]^T = [10, 2]^T$$

A useful second interpretation is to view Av as a linear combination of the columns of A . If the columns are c_1 and c_2 , then

$$A[x, y]^T = x c_1 + y c_2$$

This column-combination viewpoint is often more insightful than mechanical row multiplication. The input coordinates tell the matrix how much of each transformed basis vector to combine.

Linearity

Matrix multiplication is linear. For vectors u and v and scalars α and β ,

$$A(\alpha u + \beta v) = \alpha Au + \beta Av$$

This is the algebraic reason that knowing the action of A on a basis determines its action on every vector.

4. Matrices as linear transformations

A matrix is more than a table of numbers. It represents a transformation of an entire vector space. Every input vector is moved according to the same linear rule.

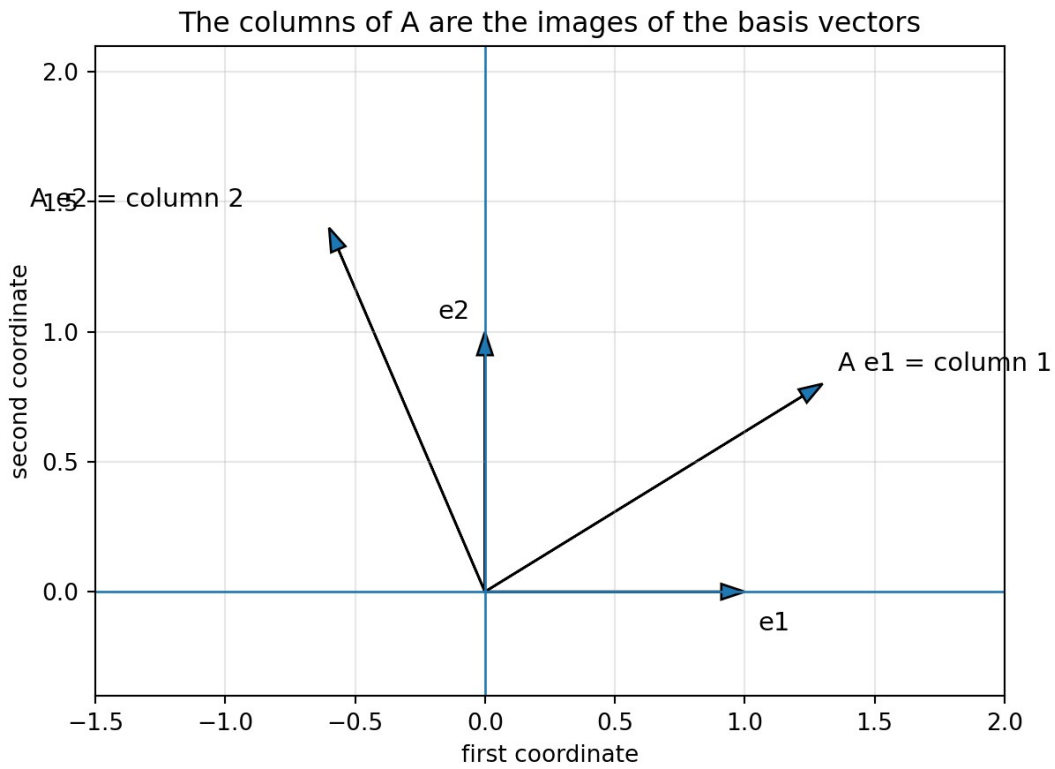


Figure 3. The columns of a matrix show where the standard basis vectors are sent.

$A e_1 = \text{first column of } A, \quad A e_2 = \text{second column of } A$

For a two-dimensional vector $v = x e_1 + y e_2$, linearity gives

$$Av = x A e_1 + y A e_2$$

Thus, the matrix columns are not arbitrary storage locations. They are the transformed basis vectors from which every output is constructed.

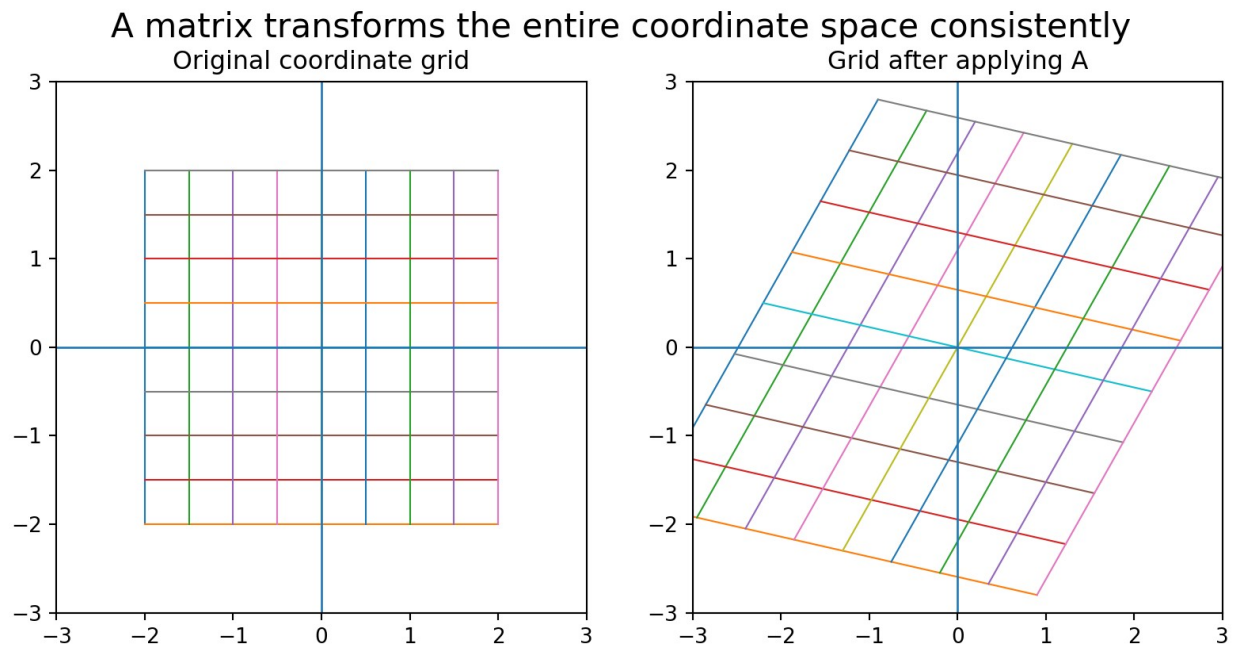


Figure 4. A linear transformation changes the coordinate grid while preserving straight lines and the origin.

What linear transformations preserve

- The zero vector maps to the zero vector: $A0 = 0$.
- Lines through the origin remain lines through the origin, although they may rotate, stretch, reflect, shear, or collapse.
- Parallel combinations and ratios along a line are preserved.
- Curved or translated transformations are not represented by an ordinary matrix alone.

5. Matrix addition and scalar multiplication

Matrices with identical dimensions can be added component by component. Scalar multiplication multiplies every entry by the scalar.

$$[[a,b],[c,d]] + [[e,f],[g,h]] = [[a+e,b+f],[c+g,d+h]]$$

$$k[[a,b],[c,d]] = [[ka,kb],[kc,kd]]$$

These operations make the set of $m \times n$ matrices into a vector space. Matrix addition is commutative and associative, but matrix multiplication behaves differently.

Linear combinations of operators

Expressions such as $aI + bX$ or $\alpha A + \beta B$ are linear combinations of matrices. Such combinations appear throughout quantum mechanics. For example, a general one-qubit Hermitian observable can be expressed using the identity and Pauli matrices.

Important distinction: A linear combination of valid quantum gates is not automatically a valid gate. It must still satisfy the required structural conditions, especially unitarity.

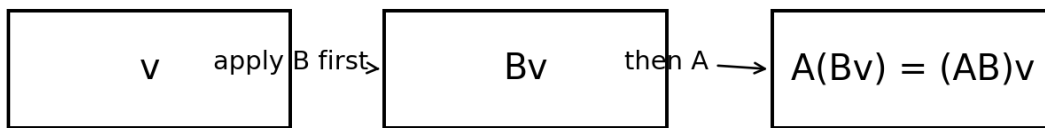
6. Matrix-matrix multiplication and composition

If A is $m \times n$ and B is $n \times p$, then AB is defined and has size $m \times p$. The shared inner dimension n must match.

$$(m \times n)(n \times p) \rightarrow (m \times p)$$

$$\begin{bmatrix} a & b \\ c & d \end{bmatrix} \begin{bmatrix} e & f \\ g & h \end{bmatrix} = \begin{bmatrix} ae+bg & af+bh \\ ce+dg & cf+dh \end{bmatrix}$$

Matrix multiplication represents composition of transformations



The rightmost matrix acts first.

Figure 5. In ABv , B acts first and A acts second.

This reversal between written order and temporal order is one of the most important conventions in quantum circuits. If a circuit applies U_1 , then U_2 , then U_3 , the final state is

$$|\psi_{\text{out}}\rangle = U_3 U_2 U_1 |\psi_{\text{in}}\rangle$$

Associativity

Matrix multiplication is associative: $(AB)C = A(BC)$. This means a long circuit can be grouped in different ways without changing the final transformation. However, multiplication is generally not commutative.

Matrix multiplication is generally not commutative: $XZ \neq ZX$

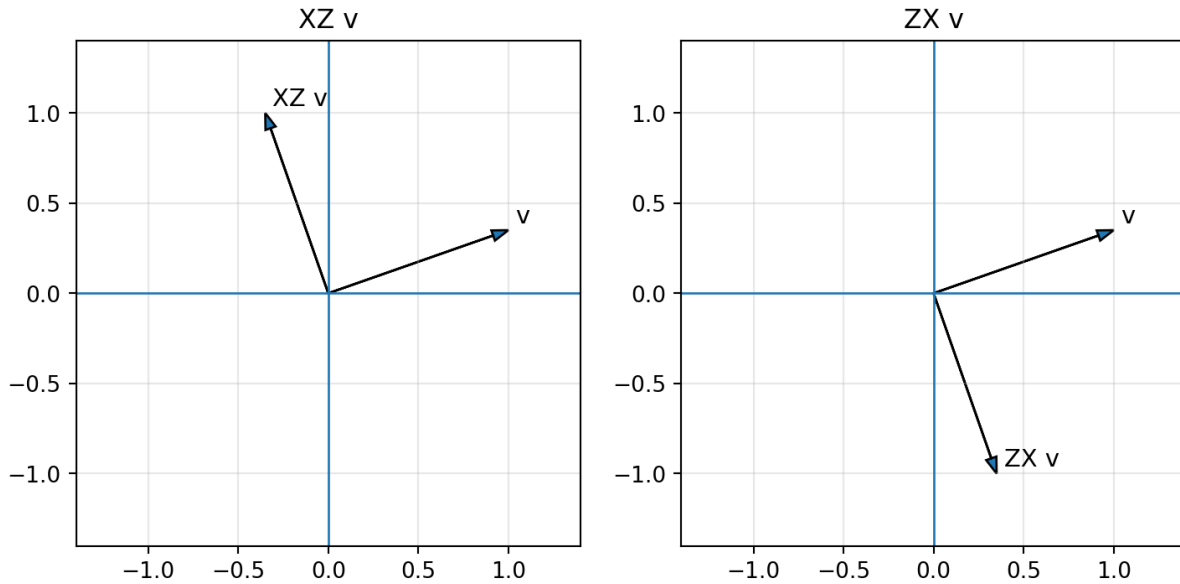


Figure 6. XZ and ZX produce different outputs for a general vector.

AB is generally not equal to BA

For Pauli matrices, $XZ = -ZX$. The minus sign may be a global phase in some isolated contexts, but operator order remains fundamentally important, especially inside larger controlled or interference-based circuits.

7. Important matrix families

Identity and zero matrices

$$I = \begin{bmatrix} 1 & 0 \\ 0 & 1 \end{bmatrix}, \quad Iv = v$$

$$0 = \begin{bmatrix} 0 & 0 \\ 0 & 0 \end{bmatrix}, \quad 0v = 0$$

The identity is the multiplicative neutral element. The zero matrix sends all vectors to zero and is maximally non-invertible.

Diagonal matrices

$$D = \text{diag}(d_1, d_2, \dots, d_n)$$

A diagonal matrix scales each standard basis direction independently. In quantum computing, diagonal gates frequently encode relative phases. For example, $Z = \text{diag}(1, -1)$, $S = \text{diag}(1, i)$, and $T = \text{diag}(1, \exp(i\pi/4))$.

Triangular matrices

Upper triangular matrices have zeros below the main diagonal; lower triangular matrices have zeros above it. Their eigenvalues are the diagonal entries, and their determinants are the product of those entries. They arise naturally in decompositions and numerical algorithms.

Permutation matrices

A permutation matrix rearranges vector components. It contains exactly one 1 in each row and each column, with all other entries zero. The X gate swaps two one-qubit amplitudes and is a simple permutation matrix.

Block matrices

Large matrices are often understood as arrays of smaller matrices. Tensor products and controlled gates are especially clear in block form. For example, a controlled operation can be viewed as applying one block when the control is 0 and another block when the control is 1.

8. Transpose, conjugate, and adjoint

Transpose

The transpose A^T exchanges rows and columns. If A is $m \times n$, then A^T is $n \times m$.

$$\text{If } A = \begin{bmatrix} a & b \\ c & d \end{bmatrix}, \text{ then } A^T = \begin{bmatrix} a & c \\ b & d \end{bmatrix}$$

Entrywise complex conjugate

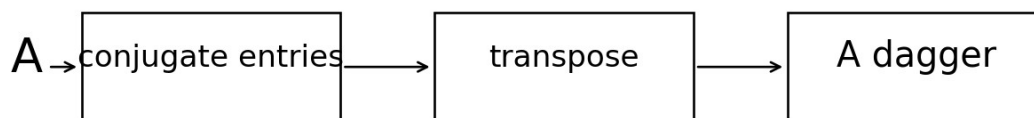
The complex conjugate A^* is obtained by conjugating every entry without changing positions.

$$\text{If } A = \begin{bmatrix} a & b \\ c & d \end{bmatrix}, \text{ then } A^* = \begin{bmatrix} a^* & b^* \\ c^* & d^* \end{bmatrix}$$

Adjoint or conjugate transpose

The adjoint $A^\dagger$ combines complex conjugation and transposition.

The adjoint (conjugate transpose)



$$\text{If } A = \begin{bmatrix} a & b \\ c & d \end{bmatrix}, \text{ then } A^\dagger = \begin{bmatrix} a^* & c^* \\ b^* & d^* \end{bmatrix}.$$

Figure 7. The adjoint is formed by conjugating entries and then transposing.

$$A^\dagger = (A^*)^T = (A^T)^*$$

For a ket $|\psi\rangle$, the adjoint produces the bra $\langle\psi|$. The adjoint is fundamental because complex inner products, unitary matrices, Hermitian matrices, and expectation values all depend on it.

Useful adjoint identities

Expression	Identity
$(A^\dagger)^\dagger$	A
$(A+B)^\dagger$	$A^\dagger + B^\dagger$
$(cA)^\dagger$	$c^* A^\dagger$
$(AB)^\dagger$	$B^\dagger A^\dagger$
$(A^{-1})^\dagger$	$(A^\dagger)^{-1}$, when A is invertible

Order reversal: The identity $(AB)^\dagger = B^\dagger A^\dagger$ is analogous to undoing a sequence: the final operation must be undone first.

9. Determinant and invertibility

For a 2×2 matrix $A = \begin{bmatrix} a & b \\ c & d \end{bmatrix}$, the determinant is

$$\det(A) = ad - bc$$

Geometrically, the absolute determinant gives the factor by which areas are scaled. The sign records whether orientation is preserved or reversed.

The determinant measures signed area scaling in two dimensions

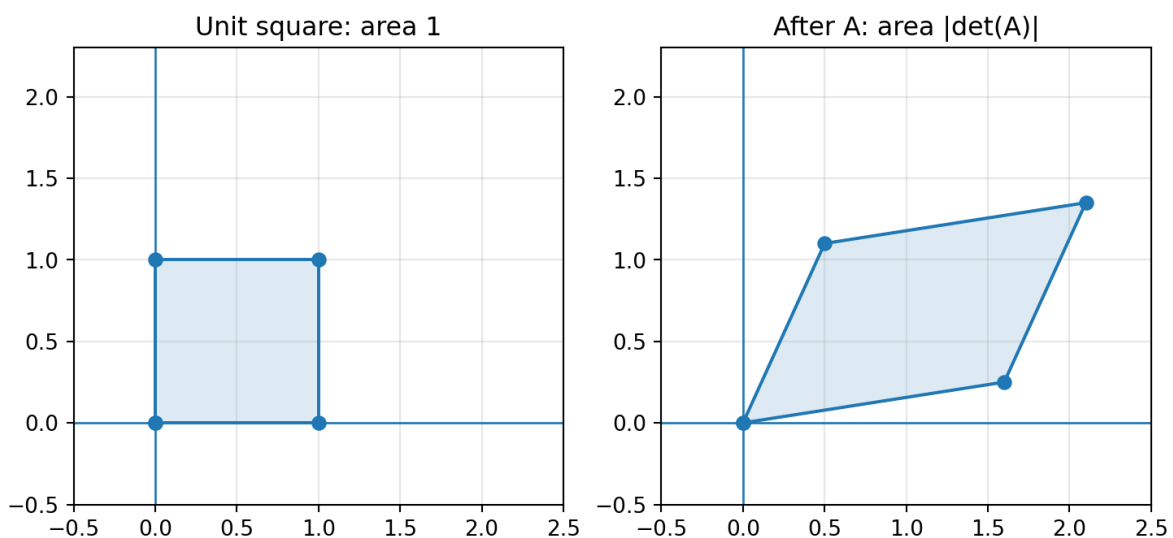


Figure 8. A nonsingular matrix maps the unit square to a parallelogram with area $|\det(A)|$.

Determinant	Geometric meaning	Algebraic consequence
$ \det(A) > 1$	Area expands	A may be invertible
$0 < \det(A) < 1$	Area contracts	A may be invertible

Determinant	Geometric meaning	Algebraic consequence
$\det(A) < 0$	Orientation reverses	A may be invertible
$\det(A) = 0$	Area collapses to zero	A is not invertible

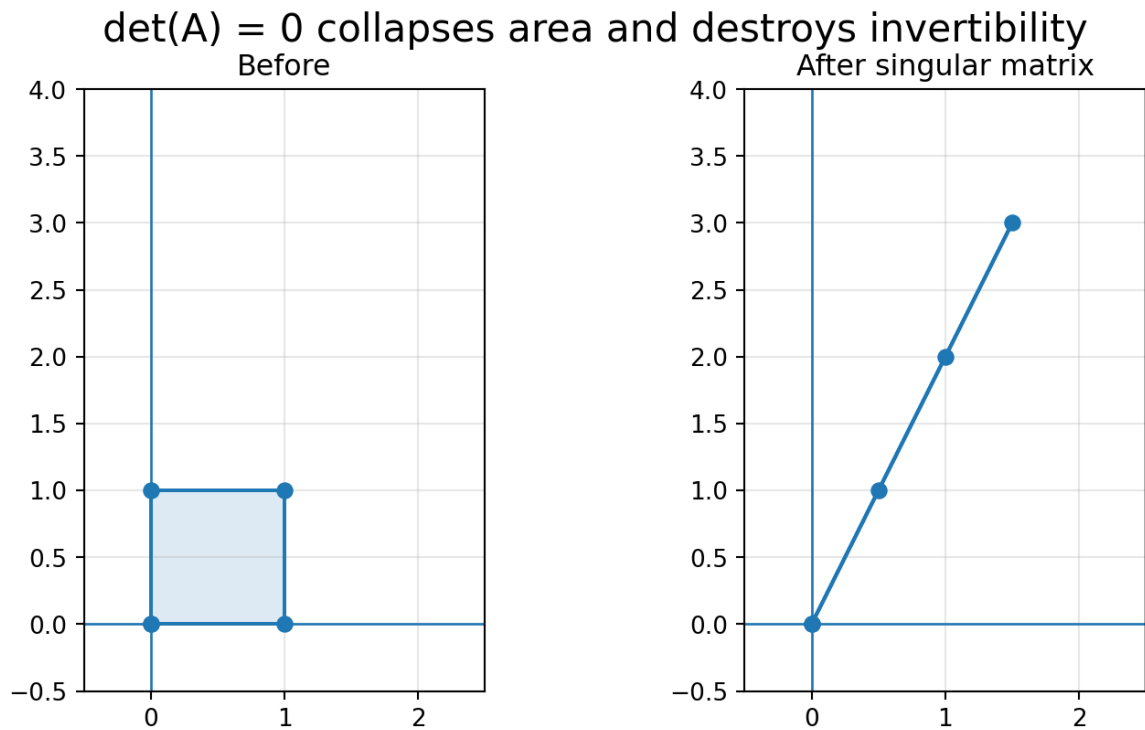


Figure 9. A singular matrix collapses a two-dimensional region into a lower-dimensional object.

Determinant properties

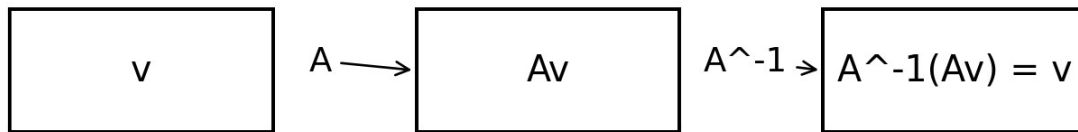
- $\det(AB) = \det(A) \det(B)$.
- $\det(I) = 1$.
- $\det(A^T) = \det(A)$.
- A is invertible exactly when $\det(A) \neq 0$.
- For a unitary matrix U, $|\det(U)| = 1$.

10. Inverse matrices and reversible transformations

An inverse matrix A^{-1} undoes A:

$$A^{-1} A = A A^{-1} = I$$

The inverse matrix undoes a reversible transformation



$$A^{-1} A = A A^{-1} = I$$

Figure 10. Applying A and then A^{-1} returns the original vector.

For a 2×2 matrix with nonzero determinant,

$$A^{-1} = (1/(ad-bc)) \begin{bmatrix} d & -b \\ -c & a \end{bmatrix}$$

Worked example

$$A = \begin{bmatrix} 2 & 1 \\ 1 & 1 \end{bmatrix}, \quad \det(A) = 2(1) - 1(1) = 1$$

$$A^{-1} = \begin{bmatrix} 1 & -1 \\ -1 & 2 \end{bmatrix}$$

Quantum gates acting on closed systems must be reversible. Their matrices are unitary, so their inverse is especially simple: $U^{-1} = U^\dagger$. Measurement, noise, and discarding subsystems are not generally described by an inverse unitary acting on the observed system alone.

Self-inverse gates: Some gates satisfy $U^2 = I$, so $U^{-1} = U$. Examples include X, Y, Z, H, and CNOT.

11. Rank, column space, and null space

Column space

The column space of A is the span of its columns. It is exactly the set of all vectors that can appear as outputs Av . Thus, the column space describes the reachable output directions.

Rank

The rank is the dimension of the column space. It counts the number of independent output directions preserved by the transformation.

Rank of a 2×2 matrix	Meaning
2	Two independent output directions; matrix is invertible

Rank of a 2 x 2 matrix	Meaning
1	Plane collapses to a line
0	Everything maps to zero

Null space or kernel

The null space is the set of vectors mapped to zero:

$$\text{Null}(A) = \{v : Av = 0\}$$

A nonzero null-space vector means two different inputs can produce the same output, because $A(u+v)=Au$ when $Av=0$. This loss of distinguishability is why a matrix with a nontrivial null space is not invertible.

Rank-nullity principle

For an $m \times n$ matrix,

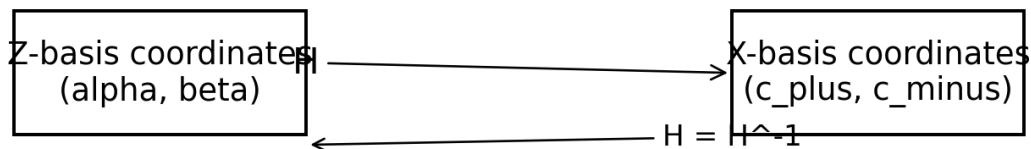
$$\text{rank}(A) + \text{nullity}(A) = n$$

The number of independent input directions is divided between directions that survive into the output and directions that are annihilated. Unitary matrices have full rank and zero nullity.

12. Basis changes and matrix representations

A physical vector or transformation does not depend on a coordinate basis, but its numerical representation does. A change-of-basis matrix converts coordinate descriptions.

Change of basis as an invertible matrix transformation



The state is unchanged; only its coordinate description changes.

Figure 11. The Hadamard matrix converts between computational-basis and X-basis coordinates.

Suppose the columns of P are the new basis vectors written in the old basis. If $[v]_{\text{new}}$ contains the new coordinates, then

$$[v]_{\text{old}} = P [v]_{\text{new}}, \quad [v]_{\text{new}} = P^{-1} [v]_{\text{old}}$$

Transforming an operator

If A_{old} represents an operator in the old basis, then its representation in the new basis is

$$A_{\text{new}} = P^{-1} A_{\text{old}} P$$

This is called a similarity transformation. If P is unitary, then $P^{-1} = P^\dagger$, so

$$A_{\text{new}} = P^\dagger A_{\text{old}} P$$

For example, $H Z H = X$. This expresses the same underlying observable in a different basis: a Z measurement after changing into the X basis corresponds to an X measurement in the original basis.

13. Quantum gates as matrices

A one-qubit state has the form $[\alpha, \beta]^T$. A one-qubit gate is a 2×2 complex matrix acting on these amplitudes. Valid closed-system gates are unitary, but this section focuses on their explicit matrix action; unitarity is studied in the special-matrices volume.

Common one-qubit matrices as transformations of amplitudes

Gate	Action on $[a, b]^T$	Main effect
I	$[a, b] \rightarrow [a, b]$	No change
X	$[a, b] \rightarrow [b, a]$	Swap basis amplitudes
Z	$[a, b] \rightarrow [a, -b]$	Relative phase flip
H	$[a, b] \rightarrow \frac{1}{\sqrt{2}}[a+b, a-b]$	Create or recombine superposition
S	$[a, b] \rightarrow [a, i b]$	Quarter-turn phase
T	$[a, b] \rightarrow [a, \exp(i \pi/4)b]$	Eighth-turn phase

Figure 12. Common one-qubit gates written as amplitude transformations.

Identity gate

$$I = \begin{bmatrix} 1 & 0 \\ 0 & 1 \end{bmatrix}, \quad I[\alpha, \beta]^T = [\alpha, \beta]^T$$

Pauli-X gate

$$X = \begin{bmatrix} 0 & 1 \\ 1 & 0 \end{bmatrix}, \quad X[\alpha, \beta]^T = [\beta, \alpha]^T$$

X swaps the amplitudes of $|0\rangle$ and $|1\rangle$. On computational-basis states it acts like a classical NOT operation.

Pauli-Z gate

$$Z = \begin{bmatrix} 1 & 0 \\ 0 & -1 \end{bmatrix}, \quad Z[\alpha, \beta]^T = [\alpha, -\beta]^T$$

Z preserves computational-basis probabilities but flips the relative phase of the $|1\rangle$ amplitude.

Hadamard gate

$$H = (1/\sqrt{2}) \begin{bmatrix} 1 & 1 \\ 1 & -1 \end{bmatrix}$$

$$H[\alpha, \beta]^T = (1/\sqrt{2})[\alpha + \beta, \alpha - \beta]^T$$

Hadamard mixes amplitudes through addition and subtraction. This is why it both creates superposition and converts relative phase into measurable interference.

Phase gates

$$S = \text{diag}(1, i), \quad T = \text{diag}(1, \exp(i\pi/4))$$

These gates leave the $|0\rangle$ component unchanged while rotating the phase of the $|1\rangle$ component. Their physical importance appears only when the qubit has coherent superposition between basis components.

Circuit composition example

Starting from $|0\rangle$, apply H, then Z, then H:

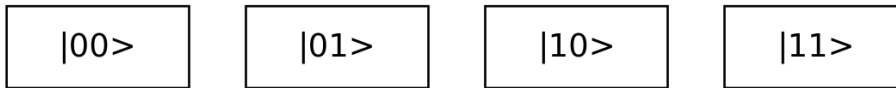
$$H Z H |0\rangle = X |0\rangle = |1\rangle$$

The matrix product HZH equals X. Algebraic matrix identities therefore translate directly into circuit equivalences.

14. Multi-qubit matrices and tensor products

A two-qubit state is a four-component complex vector

Two-qubit computational basis



General state $\alpha_{00}|00\rangle + \alpha_{01}|01\rangle + \alpha_{10}|10\rangle + \alpha_{11}|11\rangle$

State vector $[\alpha_{00}, \alpha_{01}, \alpha_{10}, \alpha_{11}]^T$

Figure 13. Two qubits require four complex amplitudes.

The basis ordering convention must be stated. A common ordering is $|00\rangle, |01\rangle, |10\rangle, |11\rangle$. A two-qubit state is then a four-component vector in that order.

Tensor product of state vectors

$$[a, b]^T \text{ tensor } [c, d]^T = [ac, ad, bc, bd]^T$$

For example, $|0\rangle \text{ tensor } |1\rangle = |01\rangle$. Tensor products combine separate subsystem descriptions into the joint-system vector space.

Tensor product of matrices

Kronecker (tensor) product of matrices

$$A = \begin{bmatrix} a & b \\ c & d \end{bmatrix} \quad B = \begin{bmatrix} e & f \\ g & h \end{bmatrix}$$

$$A \text{ tensor } B = \begin{bmatrix} aB & bB \\ cB & dB \end{bmatrix}$$

Tensor products build operators on combined quantum systems.

Figure 14. A tensor product replaces each entry of A by a scaled copy of B.

If A is $m \times n$ and B is $p \times q$, then A tensor B is $(mp) \times (nq)$

To apply X only to the second qubit, use I tensor X. To apply X only to the first qubit, use X tensor I. The apparent left-right convention depends on the chosen basis ordering, so consistency is essential.

Worked example: I tensor X

I tensor X maps $|00\rangle \rightarrow |01\rangle$, $|01\rangle \rightarrow |00\rangle$, $|10\rangle \rightarrow |11\rangle$, $|11\rangle \rightarrow |10\rangle$

CNOT

CNOT as a permutation matrix on the computational basis



The control qubit flips the target only when the control is 1.

Figure 15. CNOT permutes basis states according to a conditional rule.

$$\text{CNOT} = \begin{bmatrix} 1 & 0 & 0 & 0 \\ 0 & 1 & 0 & 0 \\ 0 & 0 & 0 & 1 \\ 0 & 0 & 1 & 0 \end{bmatrix}$$

CNOT is not equal to a tensor product of two independent one-qubit gates. It couples the qubits and can create entanglement. For example, applying H to the first qubit and then CNOT to $|00\rangle$ produces $(|00\rangle + |11\rangle)/\sqrt{2}$.

15. Worked examples

Example 1: infer a matrix from basis actions

Suppose $A e_1 = [2, 1]^T$ and $A e_2 = [-1, 3]^T$. The columns of A are these two output vectors, so

$$A = \begin{bmatrix} 2 & -1 \\ 1 & 3 \end{bmatrix}$$

Example 2: compose two gates

Apply X first and H second. The total operator is HX:

$$HX = (1/\sqrt{2}) \begin{bmatrix} 1 & 1 \\ 1 & -1 \end{bmatrix}$$

Applied to $|0\rangle$, X produces $|1\rangle$, and H then produces $|-\rangle$. The matrix product gives the same result.

Example 3: identify singularity

$$A = \begin{bmatrix} 1 & 2 \\ 2 & 4 \end{bmatrix}, \quad \det(A) = 4 - 4 = 0$$

The second row and column directions are dependent. The matrix has rank 1, a nontrivial null space, and no inverse.

Example 4: adjoint of a complex matrix

$$A = \begin{bmatrix} 1 & i \\ 2-i & 3 \end{bmatrix}$$

$$A^\dagger = \begin{bmatrix} 1 & 2+i \\ -i & 3 \end{bmatrix}$$

Example 5: one-qubit gate on an arbitrary state

$$H(\alpha|0\rangle + \beta|1\rangle) = ((\alpha+\beta)/\sqrt{2})|0\rangle + ((\alpha-\beta)/\sqrt{2})|1\rangle$$

This expression displays interference explicitly: amplitudes associated with the same output are summed before probabilities are calculated.

16. Common mistakes and diagnostic checks

Mistake	Why it fails	Correction
Multiplying entries position by position	That is the Hadamard/elementwise product, not matrix multiplication	Use row-by-column multiplication
Reading dimensions backward	Matrix dimensions are rows x columns	Check inner dimensions before multiplying
Applying gates left to right in a product	In $U_2 U_1 \psi\rangle$, U_1 acts first	Read operator products from right to left
Assuming $AB = BA$	Most transformations do not commute	Test both products or use known commutation rules
Using transpose instead of adjoint	Complex conjugation is required in complex spaces	Use $A^\dagger$, not merely A^T
Treating $\det(A)$ as the total scaling of vector length	The determinant scales volume, not every individual length	Use singular values/norms for directional length scaling
Ignoring basis order in multi-qubit matrices	The same numeric matrix can be interpreted differently	State and maintain a basis-order convention
Assuming every matrix is a valid quantum gate	Closed-system gates must be unitary	Check $U^\dagger U = I$

Fast sanity check: For a one-qubit gate, confirm a 2×2 matrix acts on a two-entry vector; confirm the output remains two entries; and, for a physical gate, confirm normalization is preserved.

17. Exercises**A. Matrix notation and basic operations**

16. For $A = \begin{bmatrix} 2 & -1 & 3 \\ 0 & 4 & 5 \end{bmatrix}$, state the dimensions and identify a_{23} .

17. Compute $[[1,2],[3,4]] + [[-1,5],[2,0]]$.
18. Compute $(2-i)[[1,i],[-1,2]]$.
19. Compute $[[2,1],[-1,3]][4,2]^T$.
20. Express $A[x,y]^T$ as a linear combination of the columns of A .

B. Products and composition

21. Compute AB and BA for $A=[[1,1],[0,1]]$ and $B=[[1,0],[1,1]]$.
22. Given a circuit that applies H , then S , then X , write the total operator in the correct order.
23. Verify directly that $X^2=I$ and $H^2=I$.
24. Calculate HZH and identify the resulting familiar gate.
25. Explain why $(AB)^\dagger = B^\dagger A^\dagger$ is reasonable from the viewpoint of reversing operations.

C. Determinant, inverse, rank, and null space

26. Find $\det([[3,1],[2,1]])$ and determine whether the matrix is invertible.
27. Find the inverse of $[[2,1],[1,1]]$.
28. Find a nonzero vector in the null space of $[[1,2],[2,4]]$.
29. Determine the rank of $[[1,2],[2,4]]$.
30. Explain geometrically what a zero determinant means.

D. Quantum matrices

31. Apply X , Z , and H separately to the state $[\alpha, \beta]^T$.
32. Apply Z to $|+\rangle$ and identify the result.
33. Apply H to $|-\rangle$ and identify the result.
34. Calculate $S^\dagger$ and verify $S^\dagger S = I$.
35. Using the basis order $|00\rangle, |01\rangle, |10\rangle, |11\rangle$, write the vector for $(|00\rangle + i|11\rangle)/\sqrt{2}$.
36. Compute $(I \otimes X)|10\rangle$.
37. Compute $CNOT(H \otimes I)|00\rangle$.

E. Conceptual questions

38. Why do the columns of a matrix completely determine a linear transformation?
39. Why can a singular matrix not represent a reversible closed-system quantum gate?
40. What is the distinction between a vector being changed and its coordinates being changed by a basis transformation?
41. Why does order matter in quantum circuits?
42. Why is $CNOT$ not an independent action on each qubit?

18. Complete solutions

A. Matrix notation and basic operations

1. The matrix has 2 rows and 3 columns, so it is 2×3 . The entry a_{23} is 5.
2. The sum is $[[0,7],[5,4]]$.
3. Multiply every entry by $2-i$: $[[2-i, 1+2i],[-2+i, 4-2i]]$.
4. The product is $[10,2]^T$.
5. If A has columns c_1 and c_2 , then $A[x,y]^T = x c_1 + y c_2$.

B. Products and composition

6. $AB = [[2,1],[1,1]]$, while $BA = [[1,1],[1,2]]$. They are not equal.
7. The total operator is XSH because H acts first, S second, and X last.
8. Direct multiplication gives $X^2=I$ and $H^2=I$.
9. $HZH=X$. This can be verified by direct multiplication or by tracking the basis action.
10. An adjoint reverses a complex inner-product action. When a sequence AB is reversed, B must be undone or transferred first, producing $B^\dagger A^\dagger$.

C. Determinant, inverse, rank, and null space

11. $\det([[3,1],[2,1]])=3-2=1$, so the matrix is invertible.
12. The inverse of $[[2,1],[1,1]]$ is $[[1,-1],[-1,2]]$.
13. Solve $x+2y=0$. One nonzero solution is $[-2,1]^T$.
14. The rank is 1 because the second row and second column direction are dependent on the first.
15. A zero determinant means a nonzero area or volume is collapsed to zero. At least one independent direction is lost, so the map cannot be reversed.

D. Quantum matrices

16. $X[\alpha,\beta]^T=[\beta,\alpha]^T$; $Z[\alpha,\beta]^T=[\alpha,-\beta]^T$; $H[\alpha,\beta]^T=(1/\sqrt{2})[\alpha+\beta,\alpha-\beta]^T$.
17. $Z|+\rangle=|-\rangle$.
18. $H|-\rangle=|1\rangle$.
19. $S^\dagger=\text{diag}(1,-i)$, and $S^\dagger S=\text{diag}(1,(-i)i)=I$.
20. The vector is $(1/\sqrt{2})[1,0,0,i]^T$.
21. $(I \otimes X)|10\rangle=|11\rangle$.
22. $(H \otimes I)|00\rangle=(|00\rangle+|10\rangle)/\sqrt{2}$; CNOT then gives $(|00\rangle+|11\rangle)/\sqrt{2}$.

E. Conceptual questions

23. Every vector is a linear combination of basis vectors, and linearity carries the matrix action through that combination. Therefore, the basis images determine every output.
24. A singular matrix merges distinguishable input directions or annihilates one of them. Once information is lost, no inverse operation on the same space can restore it.

25. A basis change changes the numerical coordinates used to describe an unchanged abstract vector. A physical transformation changes the vector itself relative to a fixed basis.
26. Different matrices represent different physical transformations, and most matrix products are noncommutative. Changing order changes the intermediate state and therefore the final state.
27. CNOT makes the target action depend on the control state. This conditional dependence cannot be factored into one independent matrix for each qubit and allows CNOT to create entanglement.

19. Compact reference sheet

Concept	Formula or test
Matrix dimensions	$m \times n$ means m rows and n columns
Matrix-vector product	$(Av)_i = \sum_j a_{ij} v_j$
Matrix product	$(AB)_{ij} = \sum_k a_{ik} b_{kj}$
Composition order	In ABv , B acts first
Identity	$Iv = v$
Adjoint	$A^\dagger$ = conjugate transpose of A
Product adjoint	$(AB)^\dagger = B^\dagger A^\dagger$
2 x 2 determinant	$\det(A) = ad - bc$
Invertibility	A invertible iff $\det(A) \neq 0$
2 x 2 inverse	$A^{-1} = (1/\det A) \begin{bmatrix} d & -b \\ -c & a \end{bmatrix}$
Rank-nullity	$\text{rank}(A) + \text{nullity}(A) = \text{number of columns}$
Basis change	$A_{\text{new}} = P^{-1} A_{\text{old}} P$
Tensor dimensions	$(m \times n) \text{ tensor } (p \times q) \rightarrow (mp \times nq)$
One-qubit gate size	2×2
n -qubit gate size	$2^n \times 2^n$

Essential quantum gate formulas

$$X = \begin{bmatrix} 0 & 1 \\ 1 & 0 \end{bmatrix}, \quad Z = \begin{bmatrix} 1 & 0 \\ 0 & -1 \end{bmatrix}, \quad H = (1/\sqrt{2}) \begin{bmatrix} 1 & 1 \\ 1 & -1 \end{bmatrix}$$

$$S = \text{diag}(1, i), \quad T = \text{diag}(1, \exp(i\pi/4))$$

Final perspective: Matrices are best understood as linear transformations, not static arrays. Matrix entries encode how basis directions are recombined, matrix products encode sequences of operations, and multi-qubit matrices encode transformations of exponentially growing state spaces.